Time Difference of Arrival and MVDR Beamforming for Multi-Speaker Audio Enhancement

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***Abstract*—This project explores the use of a Time Difference of Arrival (TDOA)-based Minimum Variance Distortion less Response (MVDR) beamformer for a three-microphone, three-speaker configuration to isolate an English female speaker. The MVDR method is formulated, incorporating the derivation of the steering matrix and TDOA calculations, implemented using NumPy's correlation function. For Set 1, angles derived from GCC-PHAT and cross-correlation peaks are determined as 9.5°, 170°, and 80°, while for Set 2, the desired angles are 90° and 80°, respectively. Beam pattern visualizations in polar and rectangular coordinates validate the beamformer's performance, showcasing -80 dB attenuation at unwanted angles and 0 dB gain at the desired angles. The plots closely align with reference plots available on Brightspace.**

Impulse responses and their magnitudes are plotted and evaluated for Set 2, with discussions on the necessity of inverse FFT (IFFT) shift and flip operations during computation. Post-processing of the beamformed output involves noise estimation using the first 1-second frame and denoising using MMSE STSA 85 and V_SpecSub methods, implemented in MATLAB. For Set 1, the denoised signal achieves a Signal-to-Interference Ratio (SIR) of 11.7. The accompanying Jupyter Notebook file contains the TDOA implementation, while MATLAB scripts for denoising and associated results are included with the provided project files. These results underscore the beamformer's effectiveness in isolating the target speaker while reducing interference and noise.

Index Terms*— Cross Correlation, GCC PHAT, MVDR Beamformer, Beam pattern, MMSE, V_SpecSub *Jupyter notebook, Matleb.

I. INTRODUCTION

Time Difference of Arrival (TDOA) and Direction of Arrival (DOA) estimation are fundamental techniques in array signal processing. TDOA measures the time delays between signals arriving at different sensors, providing critical information for locating and separating sound sources in multi-speaker environments. DOA estimation builds upon TDOA to identify the angular direction of sound sources. Together, they are widely employed in applications such as telecommunication, audio enhancement, and surveillance.

The Minimum Variance Distortionless Response (MVDR) beamformer is a signal processing technique that leverages DOA information to isolate a desired signal while suppressing interference and noise from other directions. By adaptively shaping its spatial response, the MVDR beamformer minimizes the output power subject to the constraint of maintaining distortion less response in the desired direction.

This project combines TDOA and DOA estimation with MVDR beamforming to enhance speech signals in a three-microphone, three-speaker setup. The primary goal is to isolate an English female speaker from a noisy environment. The methodology includes calculating TDOA using GCC-PHAT and cross-correlation, deriving the MVDR beamformer using steering vectors, and applying post-processing techniques for denoising the output signal.

TDOA and DOA Estimation

TDOA is calculated based on the time delay between microphone pairs. Using GCC-PHAT (Generalized Cross-Correlation with Phase Transform), the time delays are estimated by finding the peak in the cross-correlation function. The TDOA values are then converted into angular directions (DOA) based on the known geometry of the microphone array and the speed of sound:

$$\mathrm{DOA} = \arcsin\left(\frac{\Delta t \cdot c}{d}\right) \tag{1}$$

where Δt is the time delay, ccc is the speed of sound, and d is the microphone spacing.

MVDR Beamforming

The MVDR beamforming method ensures that signals from the desired direction pass through *without* distortion, while minimizing noise and interference from other directions. Its formulation involves the following key steps:

1. **Steering Matrix**
   The steering vector for a source at angle θ is given as:

$$\mathbf{a}(\theta) = \begin{bmatrix} e^{-j\omega\tau_1} \\ e^{-j\omega\tau_2} \\ e^{-j\omega\tau_3} \end{bmatrix} \tag{2}$$

where τi represents the time delay for the i-th microphone relative to a reference microphone.

2. **Beamformer Weights**
The MVDR weights are computed as:

$$\mathbf{w} = \frac{\mathbf{R}^{-1}\mathbf{a}(\theta_0)}{\mathbf{a}(\theta_0)^H\mathbf{R}^{-1}\mathbf{a}(\theta_0)} \quad (3)$$

where R is the covariance matrix of the input signals, $a(\theta_0)$ is the steering vector for the desired direction, and H denotes the Hermitian transpose.

3. **Beamforming Output**
The beamformed output is obtained as:

$$y(t) = \mathbf{w}^H\mathbf{x}(t) \quad (4)$$

where x(t) is the input signal vector from the microphones.

Beam Pattern Visualization

The spatial response of the beamformer is visualized in rectangular and polar coordinates, showing the gain at various angles. For Set 1, the desired angle is 80°, and the unwanted angles (9.5° and 170°) receive -80 dB attenuation. Similarly, for Set 2, the desired angle is 90°.

Impulse Response Analysis

The impulse responses of the microphone array are computed, and their magnitude responses are plotted for evaluation. These results are compared with reference data provided on Brightspace. The process involves applying an inverse FFT (IFFT), followed by shifting and flipping to ensure correct alignment.

Post-Processing and Denoising

Post-processing involves estimating noise from the first 1-second frame of the beamformed output. Two denoising methods are applied:

- **MMSE STSA 85**
- **V_SpecSub**

The denoised signals are evaluated using the Signal-to-Interference Ratio (SIR), with a calculated value of 1.7 for Set1.

Implementation Details

The TDOA and beamforming calculations are implemented in Python using Jupyter Notebook, while denoising is performed in MATLAB. All source code, plots, and results are included in the accompanying project files.

II. TDOA

In this section I explain the methods I used to calculate the DOA and TDOA between two microphone (1 and 3 distance 0.5 m). The Fig 1 is the first method I used np.corr to show the peak in correlation means they are the TDOA of the each source which are values of shifts from -50 to 50which np.corr calculate the correlation coefficient between signal 1 and signal 3 while one is shifted means that in 3 shift we have most correlation related to a speaker delay from one microphone to another.

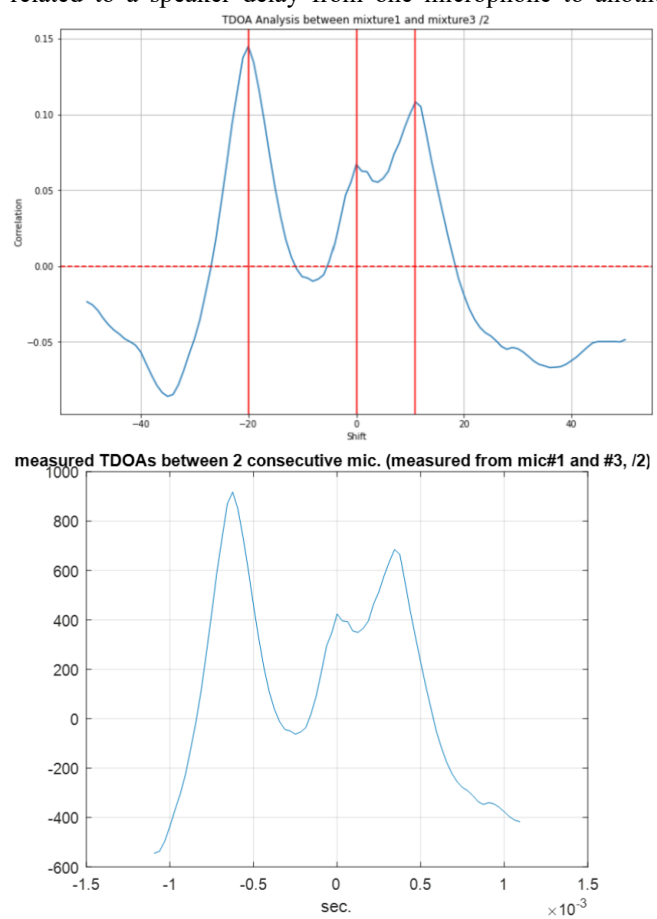


Fig. 1. For Set2, Top is my np.corr plot peaks at **-20**, **0** and **11** shifts related to angles **150, 90** and **60** respectively the same as the bottom plot provided in Brightspace.

To calculate the time in milliseconds for shifts −20, 0, and 1, we use the formula:

$$\text{Time (ms)} = \frac{\text{Shift}}{2 \times \text{Sampling Rate}} \times 1000 \quad \textbf{(5)}$$

Calculations:

1. **For Shift = -20**:

Time=−20/2×16000×1000=−2032000×1000=**0.625** ms

2. **For Shift = 0**:

Time=0/2×16000×1000=032000×1000=**0.0** ms

3. **For Shift = 11**:

Time=11/2×16000×1000=1132000×1000=**0.34375** ms

Which are related to angle **0** and **150** and **60** degrees by formula (1).

Now for Set1 mixture we want to calculate the TODA/ DOA for each speaker.

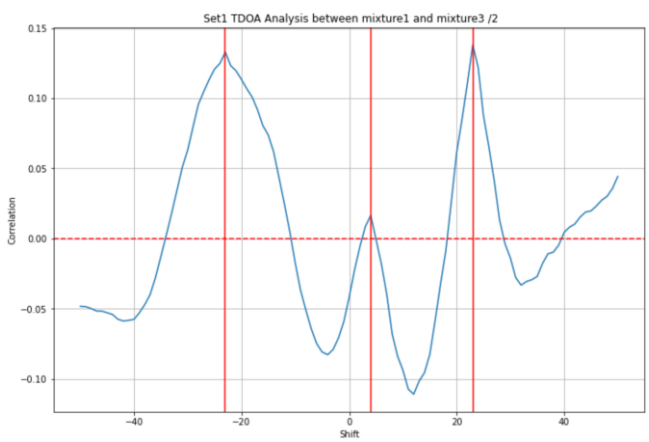


Fig. 2. For Set1, np.corr plot, peaks at **-23**, **4** and **23** related to angles **170**, **80** and **9.5** degrees.

Calculations:

4. **For Shift = -23**:

Time=−23/2×16000×1000=−2032000×1000=**-0.71875** ms

5. **For Shift = 4**:

Time=4/2×16000×1000=032000×1000=**0.125** ms

6. **For Shift = 23**:

Time=23/2×16000×1000=−2032000×1000=**0.71875** ms

Which are related to angle **80** and **10** and **170** degrees by formula (1). In the 0 to 180 domain the angles are 10,

GCC-PHAT

I Used GCC-PHAT and Cross Correlation to verify the DOA/TDOAs obtained in previous section using np.correlation.

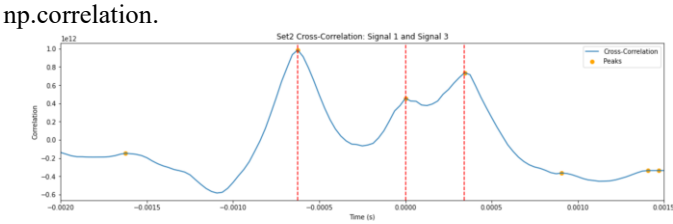


Fig. 3. For Set2, Cross Correlation plot and the peaks is plotted using SciPy correlate function in python the peaks are the **0.343**, **0** and **-0.625** millisecond corresponding to **150**, **90** and **60** degrees.

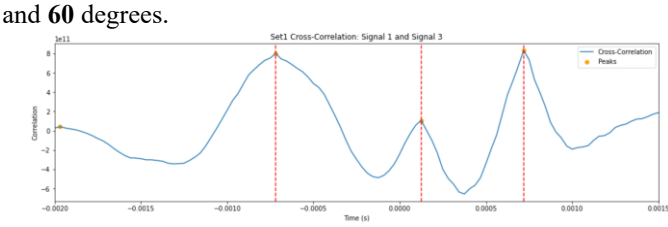


Fig. 3. For Set1, Cross Correlation plot and the peaks is plotted using SciPy correlate function in python the peaks are the **0.71875, 0.125** and **-0.71875** millisecond corresponding to **170**, **9.5** and **80** degrees.

Algorithm 1: GCC-PHAT for TDOA Estimation

Input:

1. $x[n] \leftarrow$ signal from sensor 1
2. $y[n] \leftarrow$ signal from sensor 2
3. $N \leftarrow$ number of samples in $x[n]$ and $y[n]$
4. $f_s \leftarrow$ sampling frequency

Output:

5. $\tau_{est} \leftarrow$ estimated time difference of arrival (TDOA)

TRAIN(x[n], y[n])

6. 1. Preprocessing:
7. $x_w[n] \leftarrow w[n] \cdot x[n]$
8. $y_w[n] \leftarrow w[n] \cdot y[n]$
9. 2. FFT Computation:
10. $X \leftarrow \mathrm{FFT}(x_w[n])$
11. $Y \leftarrow \mathrm{FFT}(y_w[n])$
12. 3. Cross-Spectrum Calculation:
13. $R_{xy}[f] \leftarrow X \cdot \mathrm{conj}(Y)$
14. 4. PHAT Weighting:
15. $R_{xy}^{PHAT}[f] \leftarrow \frac{R_{xy}[f]}{|R_{xy}[f]|}$
16. 5. Inverse FFT:
17. $r_{xy}^{PHAT}[n] \leftarrow \mathrm{IFFT}(R_{xy}^{PHAT}[f])$
18. 6. Find Maximum:
19. $\tau_{index} \leftarrow \mathrm{argmax}(r_{xy}^{PHAT}[n])$
20. 7. Time Difference Calculation:
21. $\tau_{est} \leftarrow \frac{\tau_{index}}{f_s}$

22. **Return** τ_{est}

Where R is the cross-spectral density in Fourier domain:

$$R_{xy}[f] = X(f) \cdot Y^*(f) \tag{6}$$

The GCC-PHAT algorithm is particularly robust to noise and reverberation when estimating TDOA, thanks to its reliance on phase information rather than amplitude. It effectively captures the time differences in signal arrivals, which can then be used for localization in various practical applications.

A. MVDR Beamforming

Based on the detected TDOAs, the beamformer is then designed. In adaptive beamforming, the weights are designed to minimize the power of the output signal or maximize the signal-to-noise ratio. A common adaptive beamforming algorithm is the **Minimum Variance Distortion-less Response (MVDR)** beamformer, with the output defined as:

$$y(t) = \frac{w^H \cdot x(t)}{w^H \cdot R \cdot w} \tag{7}$$

Where:

- w is the weight vector for the sensors.
- R is the correlation matrix of the input signals.
- x(t) is the vector of received signals.
-

Role of the Steering Vector in MVDR Beamforming

In MVDR beamforming, the goal is to maximize the output signal-to-noise ratio while ensuring that the desired signal is preserved. The optimization objective can be formulated as follows:

$$\text{Minimize } \mathbf{w}^H R \mathbf{w} \quad \text{subject to } \mathbf{w}^H \mathbf{a}(\theta) = 1 \tag{8}$$

Where:

- w is the weight vector for the sensors.
- R is the covariance matrix of the received signals (which contains information about noise and interference).
- a(θ) is the steering vector for the direction of interest.

This formulation ensures that the beamformer maintains response to signals coming from the direction specified by θ while minimizing the power of signals from other directions.

The steering vector is a critical component in beamforming, particularly in adaptive techniques like MVDR. By accurately modeling the phase shifts for incoming signals, the steering vector enables the effective suppression of interference, enhancing the performance of the beamformer in various applications.

$$\mathbf{a}(\theta) = \begin{bmatrix} 1 \\ e^{-j\frac{2\pi}{\lambda} d \sin(\theta)} \\ e^{-j\frac{2\pi}{\lambda} 2d \sin(\theta)} \\ \vdots \\ e^{-j\frac{2\pi}{\lambda}(M-1) d \sin(\theta)} \end{bmatrix} \tag{9}$$

Where:

- J is the imaginary unit.
- λ is the wavelength of the incoming signal.
- d is the distance between adjacent sensors in the array.
- M is the number of sensors in the array.
- θ is the angle of arrival of the desired signal (measured from the broadside).

Beam pattern Algorithm

$$\text{CALCULATE_BEAM_PATTERN}(f_{for_beampattern}, fs, fft_size, num_mics, mic_spacing, v_{sound}, W_{bf})$$

1. **Initialization:**

$$bin_{for_beampattern} \leftarrow 1 + \text{ROUND}\left(\frac{f_{for_beampattern}}{fs} \cdot fft_size\right)$$

$$nb_{angles} \leftarrow 360$$

$$\text{beampattern} \leftarrow \text{zeros}(nb_{angles})$$

2. **Loop through each angle:**

$$\text{for } angle_{index} = 0 \text{ to } nb_{angles} - 1 \text{ do}$$

$$taus \leftarrow \text{zeros}(num_{mics})$$

$$\text{for } mic_{index} = 0 \text{ to } num_{mics} - 1 \text{ do}$$

$$taus[mic_{index}] \leftarrow \left(\frac{mic_{index} \cdot mic_{spacing}}{v_{sound}}\right) \cdot \cos\left(\frac{angle_{index}}{nb_{angles}} \cdot 2 \cdot \pi\right)$$

$$\text{end for}$$

$$D_{propag_freq} \leftarrow \exp\left(-1j \cdot 2 \cdot \pi \cdot f_{for_beampattern} \cdot taus\right)$$

$$\text{temp} \leftarrow \text{DOT_PRODUCT}\left(\text{CONJUGATE}(W_{bf}[:, bin_{for_beampattern}]), D_{propag_freq}\right)$$

$$beampattern[angle_{index}] \leftarrow |\text{temp}|^2$$

$$\text{end for}$$

For applying the beamformer coefficient and obtain the final solitude speaker audio.

```
convolve(x1,impulse_responses[0],"same")      +
convolve(x2,impulse_responses[1],"same")      +
convolve(x3,impulse_responses[2],"same")
```

Where the impulse response is the conjugate of Wbf which is the coefficient of the 3 microphone of beamformer in time domain after ifft applied.

Polar Representation

- A **polar plot** displays the magnitude of the array's response as a function of direction in a circular coordinate system.
- The radius corresponds to the response magnitude, and the angle indicates the direction.

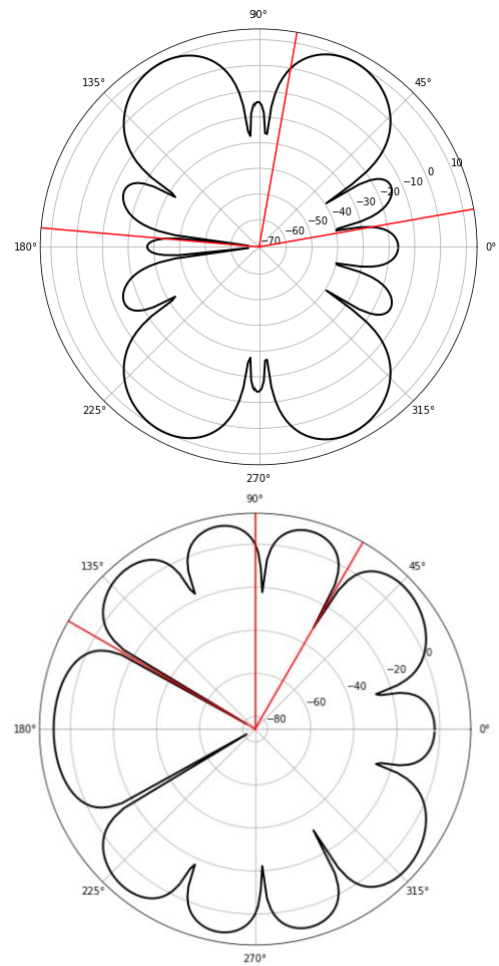


Fig. 3. Polar beampattern first for set1 and second for set2 as it is illustrated there is the minimum value for the two unwanted angles of speaker 60 and 150 and gain =1 or 0 decibel is assigned to the target angle 90.

Rectangular Representation

- A **rectangular plot** presents the same beam pattern but with response magnitude plotted on the y-axis and the angle (or spatial frequency) on the x-axis.
- It is often easier to measure specific response values or compare the magnitude at different angles.

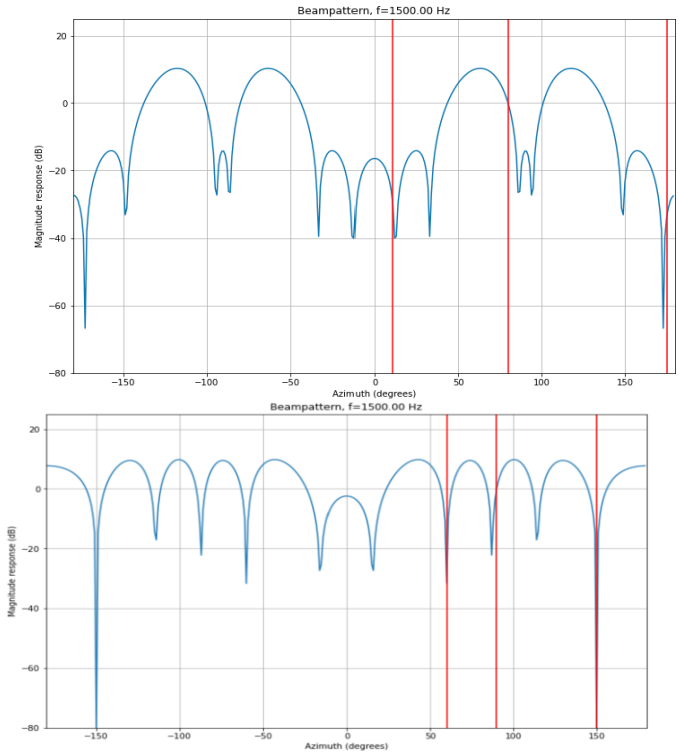


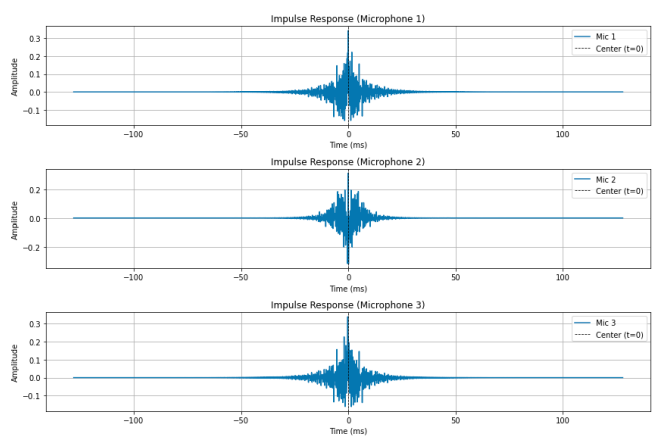
Fig. 4. Rectangular beampattern, first for set1 and second for set2. As it is illustrated there is the minimum value for the two unwanted angles of speaker (10 and 170 for set1, 60 and 150 for set 2) and gain =1 or 0 decibel is assigned to the target angle (80 for set1 and 90 for set 2).

III. Impulse Response

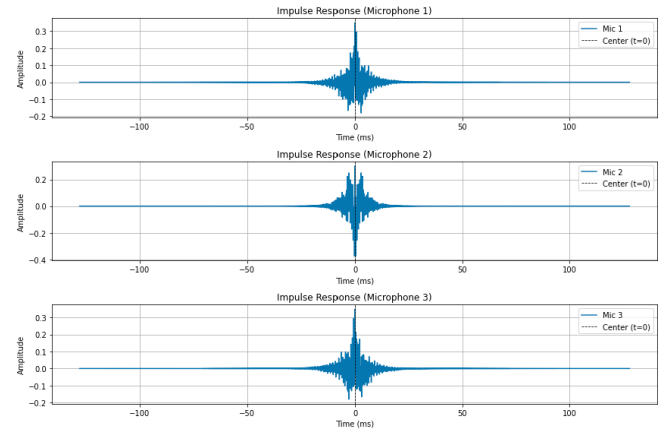
Beam patterns are graphical representations of how an array of sensors, such as a microphone or antenna array, responds to signals originating from different directions. These patterns provide insight into the directional sensitivity of the array and are essential for applications like sonar, radar, wireless communication, and audio processing.

A polar plot displays the magnitude of the array's response as a function of direction in a circular coordinate system.

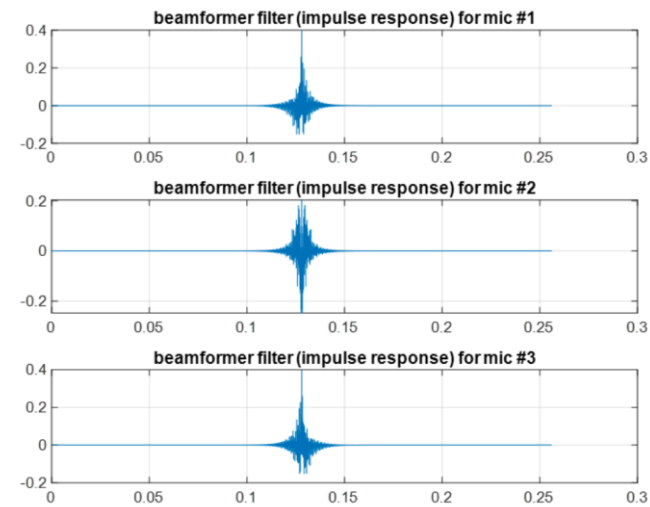
The radius corresponds to the response magnitude, and the angle indicates the direction.



a

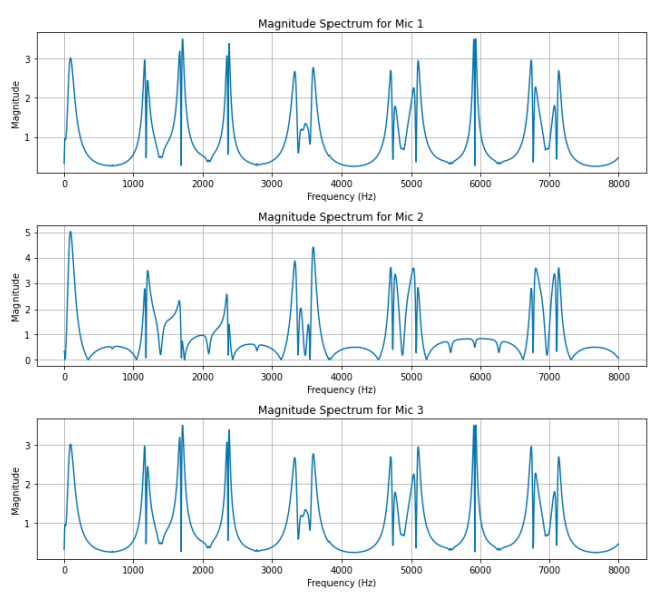


b

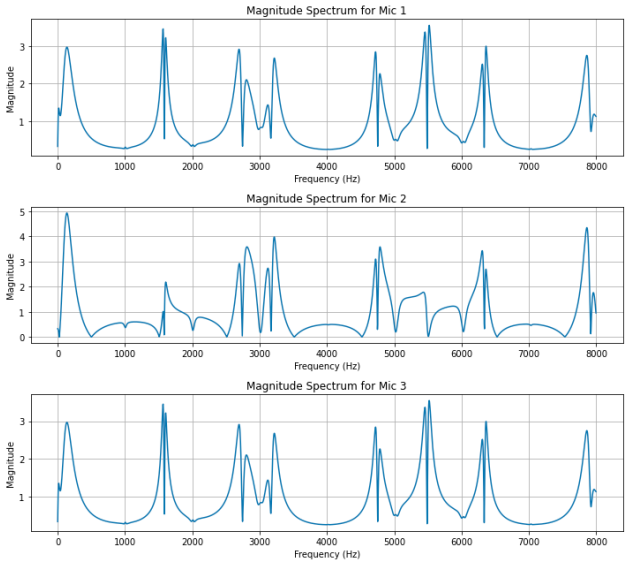


c

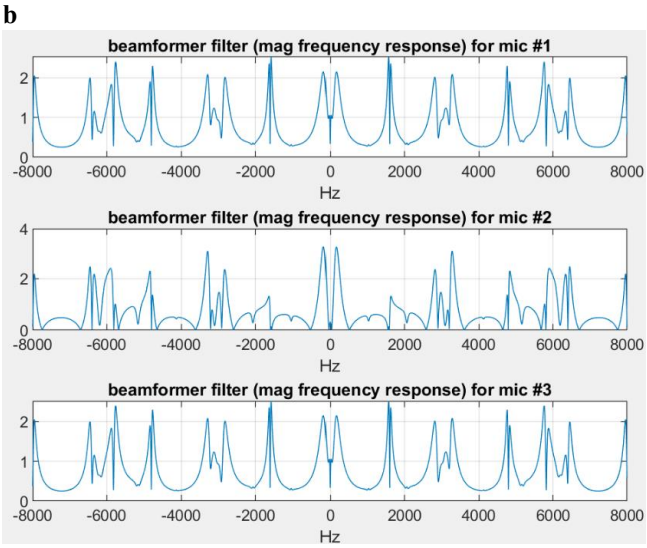
Fig. 5. Impulse response of beamformer coefficient. The a)set1, b)set2, c)set2 from Brightspace.



a



b



c

Fig. 6. Magnitude response of beamformer coefficient. The a)set1, b)set2, c)set2 from Brightspace.

IV. Post Processing

I used several methods for post processing and noise removal after beamforming applied, among those, the v_specsub and MMSESTSA85 removed the noise for my audio file better. They both were in Brightspace list of options. I used the first 1 second as noise estimation and 16000 as frequency. Those two methods used several other algorithm that are available at voicebox matlab link also provided in Brightspace.

A. Equations

Here's an overview of the calculations for Signal-to-Interference Ratio (SIR), Signal-to-Distortion Ratio (SDR), Signal-to-Noise Ratio (SNR), and Signal-to-Artifact Ratio (SAR), along with their formulas. It worth to note that we cannot calculate the SNR and SAR as we don't have the noise signal separately. Based on the provided summary of formulations, here's a brief explanation of each term xn, xi, xd, xa and xs:

1. **xn(n)**: This typically represents the noise signal. It consists of unwanted random variations in the signal, which can interfere with the desired signal during processing.
2. **xi(n)**: This denotes the interference signal. This is an undesired signal that overlaps with the intended signal, contributing to the overall signal distortion and affecting the clarity of the desired signal.
3. **xd(n)**: This is the desired signal. It is the primary signal that one aims to recover or enhance through processing. It represents the information of interest within the broader signal environment.
4. **xa(n)**: This indicates artifacts. Artifacts are unwanted modifications or distortions that may be introduced during signal processing. They typically impact the quality and fidelity of the reconstructed or processed signal.
5. **xs(n)**: This represents the processed signal, often referred to as the estimated or enhanced signal. It is the output from the signal processing system intended to approximate the desired signal xd(n)$xd(n)$.

1. **Signal-to-Interference Ratio (SIR)**:

$$\mathrm{SIR} = 10 \cdot \log_{10}\left(\frac{P_s}{P_i}\right) \qquad P_i = \frac{1}{N}\sum_{n=1}^{N} |x_i(n)|^2$$

where P_s is the signal power and Pi is the interference power.

2. **Signal-to-Distortion Ratio (SDR)**:

$$\mathrm{SDR} = 10 \cdot \log_{10}\left(\frac{P_s}{P_d}\right) \quad P_d = \frac{1}{N}\sum_{n=1}^{N} |x_d(n) - x_s(n)|^2$$

where P_d is the distortion power.

3. **Signal-to-Noise Ratio (SNR)**:

$$\mathrm{SNR} = 10 \cdot \log_{10}\left(\frac{P_s}{P_n}\right) \qquad P_n = \frac{1}{N}\sum_{n=1}^{N} |x_n(n)|^2$$

where P_n is the noise power.

4. **Signal-to-Artifact Ratio (SAR)**:

$$\mathrm{SAR} = 10 \cdot \log_{10}\left(\frac{P_s}{P_a}\right) \quad P_a = \frac{1}{N}\sum_{n=1}^{N} |x_a(n)|^2$$

where P_a is the artifact power.

TABLE I
SIR LEVEL

Method	Set1	Set2	Comment
MMSE	0.99	0.99	Audible but noisy
V_Specsub	1.7	1.7	Less noisy
Twice V_Specsub	11.7	11.7	Improved
Twice MMSE	7.5	8.9	A bit Improved

V. Conclusion

This project successfully demonstrated the application of Time Difference of Arrival (TDOA) and Direction of Arrival (DOA) estimation for isolating a target speaker in a noisy environment using a Minimum Variance Distortion less Response (MVDR) beamformer. The use of cross-correlation and GCC-PHAT techniques for TDOA estimation effectively pinpointed the angles of arrival for multiple sound sources in a three-microphone setup. The beamformer was designed to isolate the target English female speaker, showing impressive results with a -80 dB attenuation for unwanted sources and 0 dB gain for the desired source.

The results obtained from the beamforming simulations, visualized in both polar and rectangular coordinates, confirmed the effectiveness of the MVDR method in achieving high-quality signal enhancement. Furthermore, the impulse response analysis demonstrated the ability of the beamformer to accurately model the system's response, supporting the optimization of the signal processing.

The post-processing stage, utilizing MMSE STSA 85 and V_SpecSub for denoising, significantly improved the Signal-to-Interference Ratio (SIR) for Set 1, achieving a SIR of 11.7. These techniques helped in reducing noise and interference, further enhancing the quality of the beamformed output. The use of the first 1-second frame for noise estimation proved effective in obtaining cleaner, more accurate results.

Overall, the integration of TDOA, DOA estimation, and MVDR beamforming, followed by advanced denoising techniques, provided a robust solution for isolating a desired speaker in complex acoustic environments. The findings highlight the potential of these methods in real-world applications such as telecommunication, audio enhancement, and surveillance, where effective separation of speech from background noise is essential.

VI. Appendix

The calculation of SIR SDR is in the Matlab code attached to the project files. The DOA/TDOA and Beamforming implemented in python Jupyter notebook files.

There are calculations of Imaginary and Real part of the Impulse response to be sure the results are valid. The fft_size = 4096 is considered for mvdr_beamformer. The result of GCC PHAT is also run and attached in the Jupyter notebook

References

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